PH3120 – Computational Physics
Laboratory 1
CPL106 – Fourier Transform

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01. Basics of Sinewave Signals

I. $y(t)=A\sin(2\pi Ft + \phi)$

A = Amplitude.
The peak value of the sine wave.

F = frequency.
Number of cycles per seconds.

Φ = phase.
Initial angle or horizontal shift.

II. T = period of the sinewave.

$$T = \frac{1}{F}$$

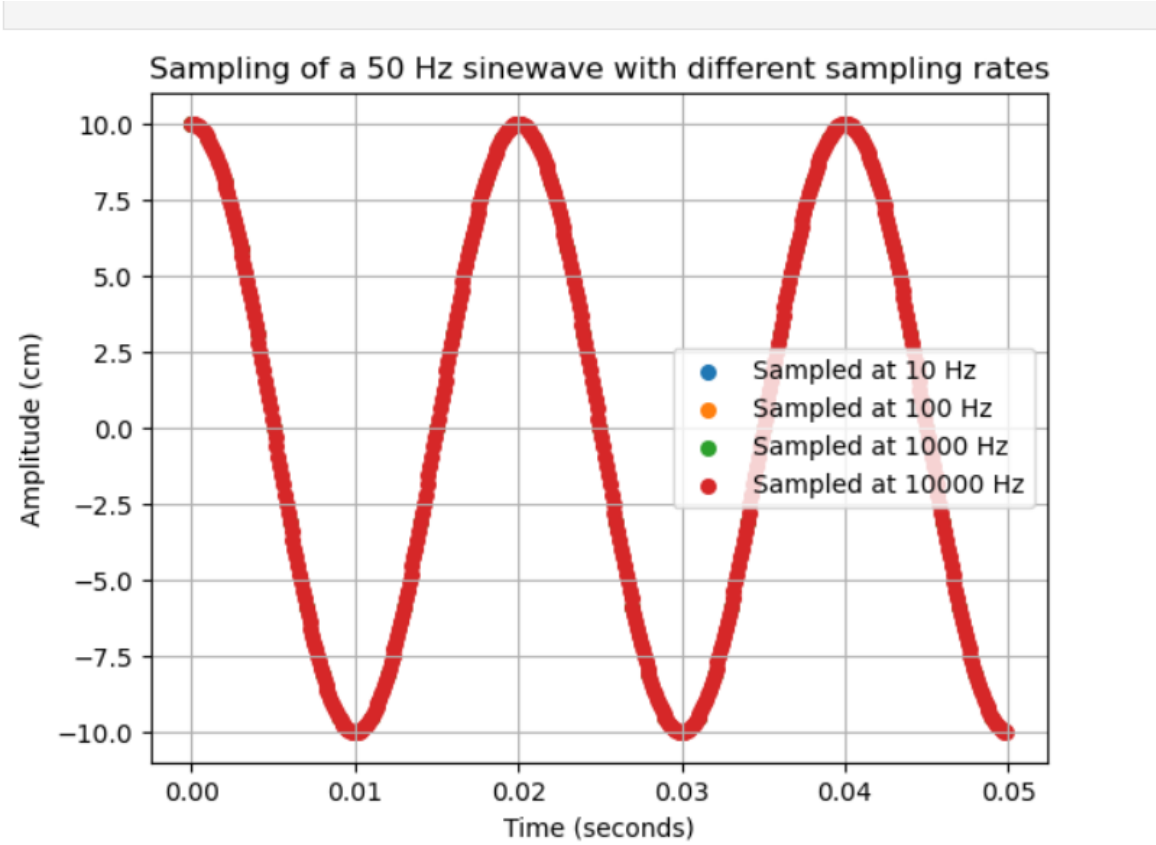
III. $T = \frac{1}{F}$

IV.

High sampling rate is important in signal sampling because it helps capture the signal more accurately. When you sample a signal more times per second, you get a better and clearer picture of how the signal changes over time. This reduces errors and distortion when the signal is converted from analog (continuous) to digital (discrete).

If the sampling rate is too low, important details can be missed, causing the signal to sound or look wrong—this is called aliasing. So, a high sampling rate ensures the digital signal closely matches the original analog signal.

V.



The above graph shows how the sampling rate effect on the sine wave signal. Sample frequency 10000 Hz is very dense and almost the overlap the continuous signal perfectly, showing a very accurate digital representation. Other Sample frequencies corresponding data points are not visible because 10000 Hz is overlapping other data points.

 Low Sampling (10 Hz**):** Since 10 Hz is much lower than the required 100 Hz (Nyquist rate) for a 50 Hz wave, the sampled points completely distort the original wave. This is called aliasing, where the sampled signal looks like a different, slower wave.

 Nyquist Rate Sampling (100 Hz): At exactly 100 Hz, we technically have enough points to reconstruct the wave, but the points are still very sparse (only two per cycle), making the actual shape hard to see clearly from just the dots.

 Higher Sampling (1000 Hz & 10000 Hz)**:** As the sampling rate increases well above the Nyquist rate, the sampled points become denser and more accurately trace the original sine wave. At 10000 Hz, the points are so close together that they almost perfectly represent the smooth continuous wave.

02. Discrete Fourier Transform (DFT)

I. The time domain is a way of representing a signal with respect to time. In this domain, we observe how a signal (such as voltage, sound, or any other quantity) changes over time.

In the time domain the x axis is time, and the y-axis shows the signal's amplitude. It shows the actual shape and variation of the signal over time. It's useful for analyzing when events occur and how the signal evolves. A signal is first sampled in the time domain, meaning we record its values at discrete time intervals.

II. The frequency domain is a way of representing a signal in terms of the frequencies it contains**,** rather than how it changes over time.

In the time domain the x axis is frequency, and the y-axis shows the signal's magnitude. It shows how much of each frequency is present in the signal and shows the spectrum (distribution of frequencies).It's useful for Understanding what kind of vibrations or patterns are in the signal and identifying things like noise.

Relation Between frequency domain and Fourier Transform :-

The Fourier Transform (and its discrete version, the Discrete Fourier Transform – DFT) is a mathematical tool that converts a signal from the time domain to the frequency domain**.** It breaks down a complex signal into a sum of simple sine and cosine waves of different frequencies.

III.

Feature	**Continuous Fourier Transform (CFT)**	**Discrete Fourier Transform (DFT)**
Input signal	Continuous in time	Discrete in time (sampled)
Output frequencies	Continuous spectrum	Discrete frequency values
Used for	Analog signals	Digital signals
Mathematical integration	Uses integrals	Uses finite summations
Domain	Time: continuous t	Time: discrete n
Output domain	Frequency: continuous f	Frequency: discrete k

(a) This is done using the Discrete Fourier Transform (DFT). For a discrete signal x[n] of length N, the DFT is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n].\, e^{-j\frac{2\pi}{N}kn} \quad \text{, k=0,1,2…,N-1}$$

Where:

- x[n] discrete time-domain signal
- X[k]: DFT output (frequency domain)
- N: total number of samples
- j: imaginary unit

(b) This is done using the Continuous Fourier Transform (CFT)**.** For a continuous-time signal x(t), the transform is:

$$X(f) = \int_{-\infty}^{\infty} x(t).\, e^{-j2\pi ft}\, dt$$

Where

- X(f): The Fourier Transform of the signal x(t) represents the signal in the frequency domain.
- x(t): The original signal in the time domain.
- f:Frequency (in Hz).
- j: Imaginary unit ($j^2 = -1$).
- $e^{-j2\pi ft}$: Complex exponential function (represents sine and cosine).

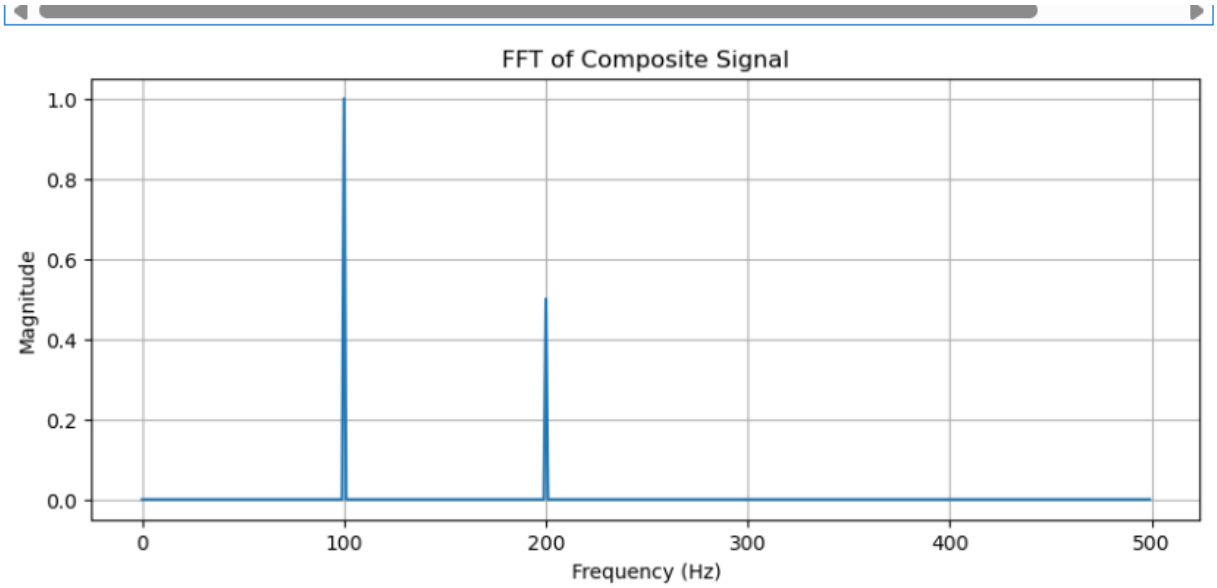
3. Fast Fourier Transform (FFT)

I. Advantages of FFT over DFT

1. Speed and Efficiency:
   FFT is significantly faster than DFT. While DFT has a computational complexity of $O(N^2)$, FFT reduces it to O(N log N). This improvement is especially important when working with large amounts of data.
2. Lower Computational Load:
   FFT requires fewer multiplications and additions compared to DFT, making it more suitable for systems with limited processing power such as embedded devices or real-time systems.
3. Real-time Processing:
   Due to its high speed, FFT can be used in real-time applications like audio signal processing, communication systems, and radar systems where quick data processing is essential.
4. Wide Availability:
   FFT algorithms are widely implemented in software libraries and hardware devices, making them readily available for use in many digital signal processing applications.

5. Memory Efficiency:
   Many FFT implementations use in-place computation techniques, which reduce the amount of memory required to perform the transform.
6. Scalability:
   FFT can handle very large data sizes efficiently, whereas DFT becomes impractical for large datasets due to its slower processing speed.

II. Done in Jupiter note book



Here is the comments on results;

The FFT plot clearly shows two distinct and sharp peaks, indicating the presence of two dominant frequency components in the original composite signal.

1. Peak 1: Frequency = 100 Hz, Magnitude = 1.0
   - This peak is located precisely at 100 Hz on the frequency axis.
   - Its magnitude is exactly 1.0. This implies that one of the constituent sine waves in the composite signal has a frequency of 100 Hz and an amplitude of 1 unit (assuming the y-axis represents normalized amplitude or voltage, depending on the original signal's units).
2. Peak 2: Frequency = 200 Hz, Magnitude = 0.5
   - This peak is located precisely at 200 Hz on the frequency axis.
   - Its magnitude is 0.5. This indicates that another constituent sine wave in the composite signal has a frequency of 200 Hz and an amplitude of 0.5 units.

Overall Observations:

- Pure Sinusoids: The extremely narrow and sharp nature of the peaks suggests that the original composite signal is formed by the superposition of very pure sine waves with precise frequencies, rather than signals with broad frequency content or significant noise.
- Absence of Other Components: There are no other significant peaks or significant "noise floor" visible across the displayed frequency range (0 to 500 Hz). This confirms that the composite signal is composed almost exclusively of these two specific sinusoidal components.
- Clarity and Resolution: The FFT successfully resolved the two distinct frequency components, demonstrating its effectiveness in spectral analysis. The x-axis extends to 500 Hz, which is likely the Nyquist frequency, meaning the original signal was sampled at least at 1000 Hz.

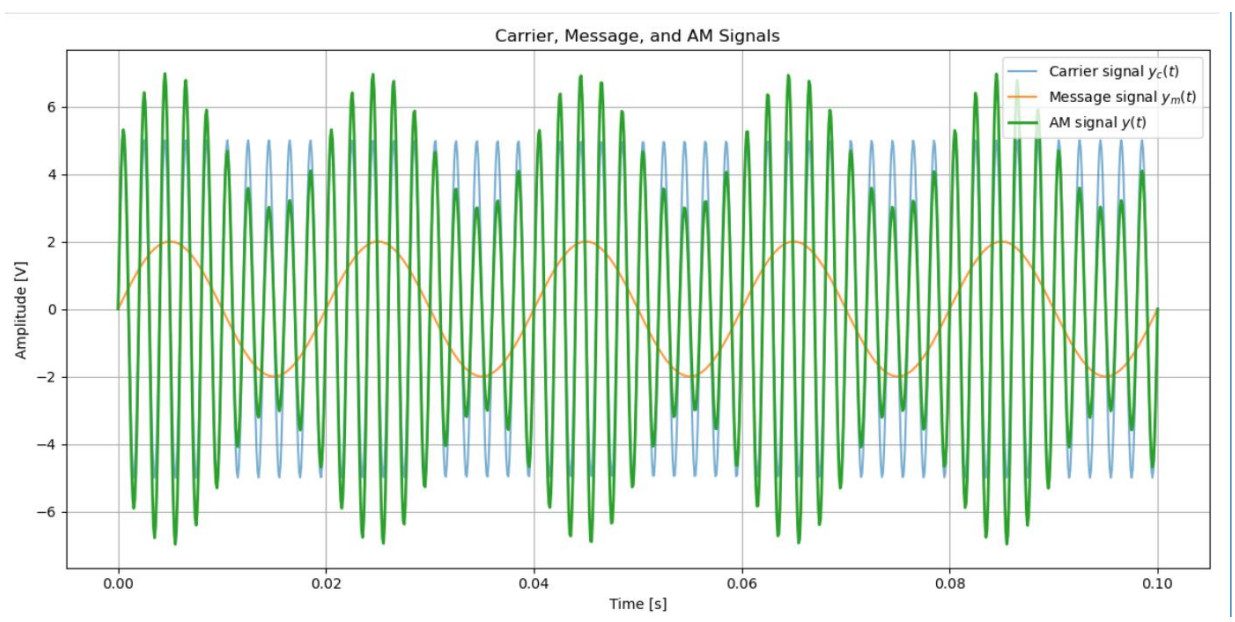
III. (a) The general form of a sine wave is $A\sin(2\pi ft)$. Given: Carrier frequency (fc) = 500 Hz, Carrier amplitude (A_c) = 5 V.
So, the carrier wave function is: $y_c(t)=A_c\sin(2\pi f_c t)=5\sin(2\pi\times 500t)$

(b) Modulating signal frequency (f_m) = 50 Hz, Modulating signal amplitude (A_m) = 2 V. So, the modulating signal function is: $y_m(t)=A_m \sin(2\pi f_m t)=2\sin(2\pi\times 50t)$

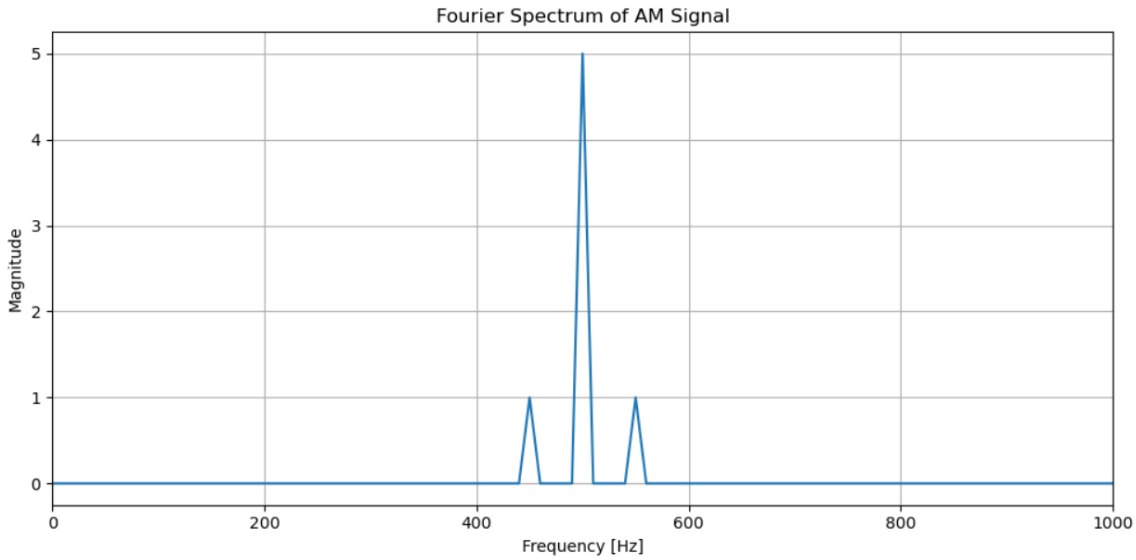
(c) In amplitude modulation, the carrier wave is multiplied by the envelope:

$$y(t) = [1 + \frac{y_m(t)}{A_m}] \cdot y_c(t)$$

(d)



(e) i.



ii. Lowest frequency in spectrum:

- 450 Hz (Lower sideband)

iii**.** Highest frequency in spectrum:

- 550 Hz (Upper sideband)

iv. Bandwidth of the signal:

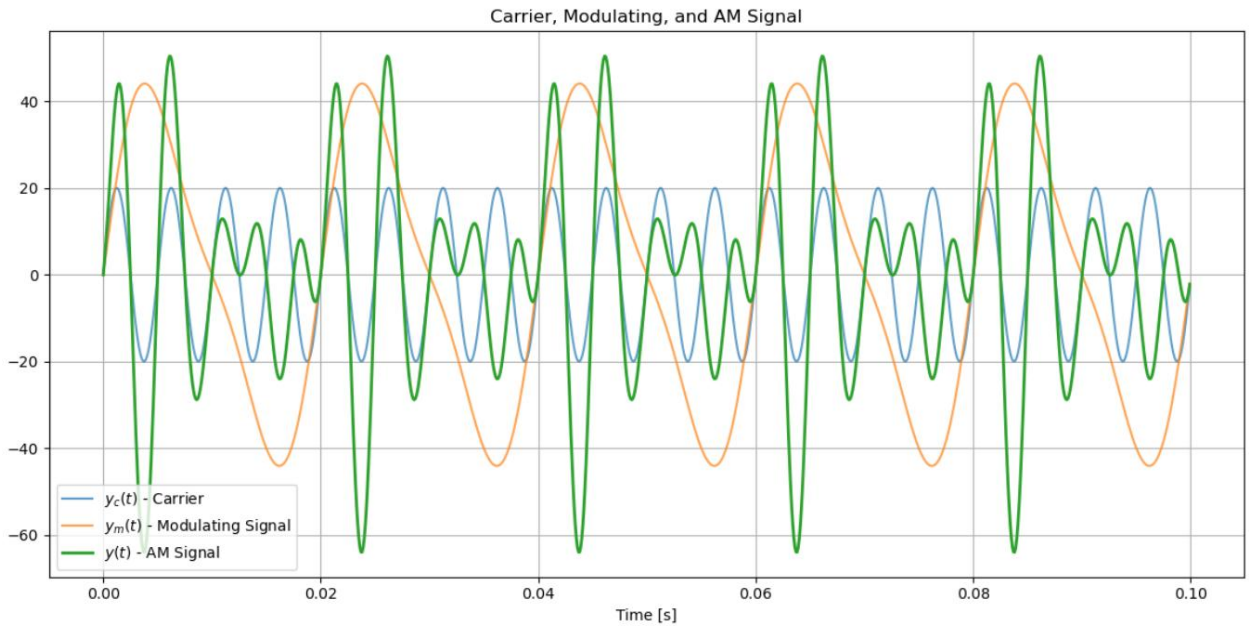
- Bandwidth =550−450=100 Hz

IV.

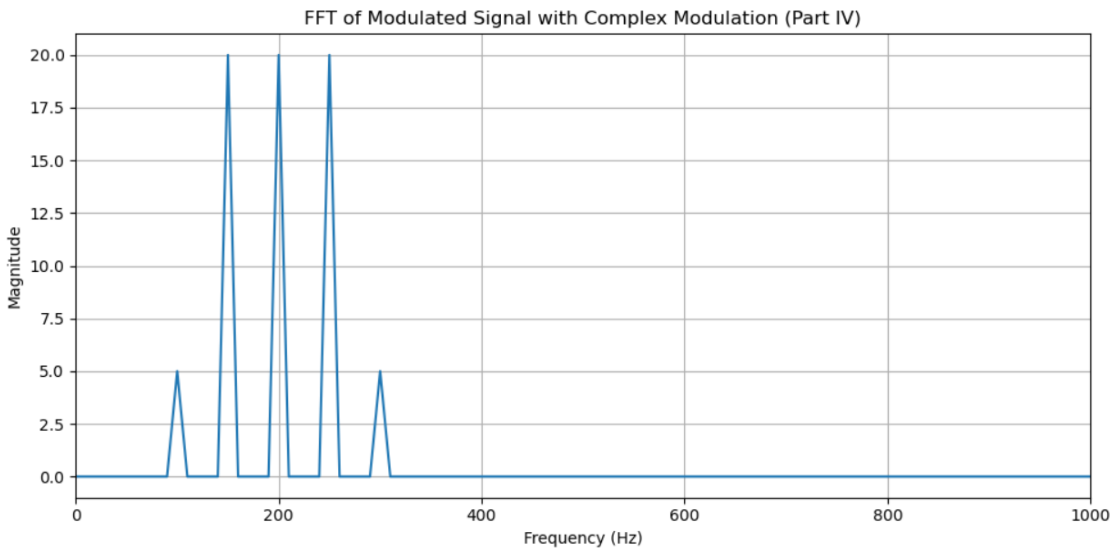
(a) $y(t) = [20 + y_m(t).\sin(2\pi.200.t))$

$$y(t) = [20 + 10\sin(200\pi t) + 40\sin(100\pi t)].\sin\,(400\pi t)$$

(b)



(c)
i.



ii. Lowest frequency in the spectrum:100Hz

iii. Highest frequency in the spectrum:300Hz

iv. Bandwidth based on the Fourier spectrum**:**200 Hz (300−100)